PC UPGRADE AND MAINTENANCE

Complete 50-Page Course Textbook

Beginner → Intermediate

Hardware • Upgrades • Maintenance • Troubleshooting • Security • Practical Exercises

A structured course covering computer hardware, safe upgrade procedures, preventive maintenance, troubleshooting, performance optimization, data protection, and practical PC servicing.

Course Overview

Purpose: This course teaches learners how to understand, maintain, troubleshoot, and safely upgrade personal computers.

Learning outcomes: Learners should be able to identify PC components, check compatibility, plan upgrades, perform preventive maintenance, diagnose common faults, protect data, maintain operating systems, and document service work.

Table of Contents

1. Introduction to PC Upgrade and Maintenance
2. Computer Hardware Fundamentals
3. Motherboard
4. CPU and Processor Basics
5. CPU Upgrade Planning
6. RAM and Memory
7. RAM Upgrade
8. Storage Devices
9. Storage Upgrade and Migration
10. Graphics Processing Unit
11. Power Supply Unit
12. PC Cooling
13. CPU Cooler Installation
14. PC Case and Airflow
15. Thermal Management
16. BIOS and UEFI
17. BIOS/UEFI Updates
18. Operating System Maintenance
19. Device Drivers
20. Disk Management
21. File Systems
22. Backup and Data Protection
23. Preventive Maintenance
24. Internal PC Cleaning
25. Electrostatic Discharge
26. Cable Management

27. PC Assembly and Disassembly
28. Compatibility Checking
29. Performance Assessment
30. Troubleshooting Fundamentals
31. PC Won't Power On
32. PC Powers On but No Display
33. Boot and Startup Problems
34. Overheating Problems
35. Slow Computer Troubleshooting
36. Storage Failure
37. Network and Peripheral Maintenance
38. Software Maintenance
39. PC Security
40. Power Protection
41. Laptop Maintenance
42. PC Upgrade Budgeting
43. Documentation and Maintenance Records
44. Professional Maintenance Workflow
45. PC Upgrade Case Study
46. Final Revision Guide
48. 60-Question Assessment and Answers
49. Practical Exercises
50. Glossary

Lesson 1: Introduction to PC Upgrade and Maintenance

Lesson Content

PC maintenance means caring for, inspecting, cleaning, updating, troubleshooting, and repairing a computer. Upgrading means adding or replacing hardware to improve capability, capacity, compatibility, or performance.

Exercise

Define upgrade, preventive maintenance, corrective maintenance, and repair.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 2: Computer Hardware Fundamentals

Lesson Content

A PC contains interconnected components such as the motherboard, CPU, RAM, storage, PSU, cooling system, case, graphics hardware, and peripherals. Each has specific functions and compatibility requirements.

Exercise

List eight major PC components and their functions.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 3: Motherboard

Lesson Content

The motherboard connects the CPU, memory, storage, expansion cards, power connectors, firmware, and peripherals. Important specifications include socket, chipset, RAM support, expansion slots, storage interfaces, and form factor.

Exercise

Why must motherboard compatibility be checked before purchasing a CPU or RAM?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 4: CPU and Processor Basics

Lesson Content

The CPU executes instructions and performs calculations. Important specifications include architecture, cores, threads, clock behavior, cache, socket compatibility, power requirements, and supported memory.

Exercise

List five CPU specifications that matter when planning an upgrade.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 5: CPU Upgrade Planning

Lesson Content

A CPU upgrade requires checking socket compatibility, motherboard firmware support, power delivery, cooling, memory compatibility, and expected performance improvement.

Exercise

Create a CPU upgrade compatibility checklist.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 6: RAM and Memory

Lesson Content

RAM is fast temporary memory used by active programs and data. Capacity, generation, speed, timings, module type, channels, and motherboard limits affect compatibility and performance.

Exercise

Explain why the wrong RAM generation may prevent a PC from starting.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 7: RAM Upgrade

Lesson Content

Before adding RAM, check supported memory type, maximum capacity, slot count, module requirements, and recommended configurations. Mixing modules can sometimes work but may reduce compatibility or performance.

Exercise

Describe a safe RAM upgrade procedure.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 8: Storage Devices

Lesson Content

HDDs use magnetic platters. SATA SSDs use flash memory through a SATA interface. NVMe SSDs normally use PCIe and offer very fast access.

Exercise

Compare HDD, SATA SSD, and NVMe SSD.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 9: Storage Upgrade and Migration

Lesson Content

Check physical dimensions, interface, motherboard support, available slots, operating-system requirements, and backups. Migration can use cloning or a clean installation.

Exercise

List the steps to take before replacing a system drive.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 10: Graphics Processing Unit

Lesson Content

A GPU handles graphics and parallel workloads. It may be integrated or dedicated. Upgrade considerations include interface, physical clearance, PSU capacity, connectors, cooling, and drivers.

Exercise

List five factors to check before installing a GPU.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 11: Power Supply Unit

Lesson Content

The PSU converts mains electricity into regulated DC power. Consider output capacity, efficiency, connectors, physical form factor, protection features, and quality.

Exercise

Why is PSU quality important during upgrades?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 12: PC Cooling

Lesson Content

Cooling removes heat from components using heatsinks, fans, airflow, and sometimes liquid cooling. Poor cooling can cause thermal throttling, instability, or reduced component life.

Exercise

Give three signs of inadequate cooling.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 13: CPU Cooler Installation

Lesson Content

A CPU cooler needs correct mounting and thermal contact. Installation involves suitable thermal interface material, secure mounting, correct fan connection, and temperature verification.

Exercise

Why is thermal interface material used?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 14: PC Case and Airflow

Lesson Content

Case selection affects component compatibility, airflow, expansion, cable management, and serviceability. Good airflow provides controlled intake and exhaust.

Exercise

Design a basic airflow plan for a desktop PC.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 15: Thermal Management

Lesson Content

High temperatures can result from dust, blocked vents, failed fans, poor cooler installation, unsuitable fan curves, or high ambient temperature.

Exercise

Give five causes of unusually high CPU temperature.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 16: BIOS and UEFI

Lesson Content

BIOS and UEFI are firmware systems that initialize hardware and provide pre-boot configuration. Modern UEFI can support secure boot and advanced device settings.

Exercise

What is the role of firmware during startup?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 17: BIOS/UEFI Updates

Lesson Content

Firmware updates may improve compatibility, stability, and security. They must be performed using the exact motherboard model and vendor instructions with stable power.

Exercise

List precautions before updating firmware.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 18: Operating System Maintenance

Lesson Content

OS maintenance includes security updates, storage cleanup, driver management, startup management, backup verification, and performance monitoring.

Exercise

Create a monthly OS maintenance checklist.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 19: Device Drivers

Lesson Content

Drivers allow operating systems to communicate with hardware. Correct drivers improve compatibility and stability. Trusted sources reduce the risk of incompatible or malicious software.

Exercise

Why should drivers come from trusted sources?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 20: Disk Management

Lesson Content

Disk management includes partitioning, formatting, storage cleanup, capacity monitoring, and understanding file systems. Back up important data before destructive operations.

Exercise

What precautions should be taken before formatting a drive?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 21: File Systems

Lesson Content

File systems organize files and directories on storage. Common PC examples include NTFS, exFAT, and FAT32, with different features and compatibility characteristics.

Exercise

Explain why file-system choice matters.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 22: Backup and Data Protection

Lesson Content

Backups protect against accidental deletion, hardware failure, corruption, malware, and other data loss. Multiple copies and separation from the main system improve resilience.

Exercise

Design a backup plan for a student's computer.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 23: Preventive Maintenance

Lesson Content

Preventive maintenance reduces failures through cleaning, ventilation checks, software updates, storage monitoring, backup verification, cable inspection, and temperature checks.

Exercise

List ten preventive maintenance tasks.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 24: Internal PC Cleaning

Lesson Content

Dust can restrict airflow and increase heat. Shut down, disconnect power, work in a suitable area, and follow manufacturer guidance when cleaning. Avoid unsafe practices that can damage components.

Exercise

Describe a safe general cleaning procedure.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 25: Electrostatic Discharge

Lesson Content

ESD can damage sensitive electronics. Use appropriate anti-static precautions, handle components by their edges, and avoid unnecessary contact with contacts.

Exercise

Why are ESD precautions important?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 26: Cable Management

Lesson Content

Good cable management improves airflow, serviceability, and organization. Keep cables away from fans and ensure connectors are seated correctly.

Exercise

State five benefits of good cable management.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 27: PC Assembly and Disassembly

Lesson Content

Assembly and disassembly should follow documentation and a logical sequence. Keep track of screws and connectors and never force components into sockets.

Exercise

Give a safe sequence for basic desktop assembly.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 28: Compatibility Checking

Lesson Content

Compatibility includes physical dimensions, interfaces, sockets, firmware, power, memory support, operating-system support, and thermal requirements.

Exercise

Create a complete PC compatibility checklist.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 29: Performance Assessment

Lesson Content

Measure performance before upgrading. Identify whether CPU, RAM, storage, GPU, thermals, networking, or software is limiting performance.

Exercise

Why should performance be measured before upgrading?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 30: Troubleshooting Fundamentals

Lesson Content

A systematic process is: identify symptoms, reproduce safely, form hypotheses, test one variable at a time, apply a fix, verify, and document.

Exercise

Write a seven-step troubleshooting method.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 31: PC Won't Power On

Lesson Content

Check mains power, power switch, PSU connections, motherboard power connectors, front-panel connections, and obvious hardware issues. Disconnect nonessential peripherals and isolate the fault.

Exercise

Create a diagnostic flow for a PC with no power.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 32: PC Powers On but No Display

Lesson Content

Possible causes include display input, cables, RAM, GPU, monitor faults, firmware settings, and hardware problems. Test systematically rather than replacing parts randomly.

Exercise

List a logical no-display troubleshooting sequence.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 33: Boot and Startup Problems

Lesson Content

Boot failures may result from storage problems, corrupted OS files, incorrect boot configuration, firmware settings, or hardware faults. Recovery tools can help isolate causes.

Exercise

Name five possible boot-failure causes.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 34: Overheating Problems

Lesson Content

Overheating can result from dust, failed fans, poor thermal contact, blocked airflow, unsuitable cooling, or heavy workloads.

Exercise

What checks should be performed when a PC overheats?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 35: Slow Computer Troubleshooting

Lesson Content

A slow PC may be caused by insufficient RAM, slow storage, high background activity, excessive startup programs, malware, thermal throttling, or outdated software.

Exercise

List six possible causes of a slow computer.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 36: Storage Failure

Lesson Content

Warning signs include errors, disappearing files, unusual crashes, slow access, and health-monitoring alerts. Important data should be backed up promptly.

Exercise

What should be done first when storage shows signs of failure?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 37: Network and Peripheral Maintenance

Lesson Content

Printers, keyboards, mice, webcams, adapters, and external drives need cable, driver, firmware, cleaning, and configuration checks. Isolate device, connection, driver, and application causes.

Exercise

Create a checklist for an unrecognized USB device.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 38: Software Maintenance

Lesson Content

Remove unused software, update trusted applications, review startup programs, maintain free storage, and avoid unknown programs.

Exercise

List five software maintenance practices.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 39: PC Security

Lesson Content

Secure maintenance includes updates, strong authentication, reputable security controls, least privilege, safe downloads, backups, and careful removable-media use.

Exercise

List eight practices that improve PC security.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 40: Power Protection

Lesson Content

Power fluctuations and outages can interrupt work and may harm equipment. Appropriate surge protection and UPS devices can provide additional protection.

Exercise

Explain the purpose of a UPS.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 41: Laptop Maintenance

Lesson Content

Laptop maintenance includes battery care, ventilation, hinges, display, keyboard, ports, storage, memory, and manufacturer-specific procedures. Upgradeability varies by model.

Exercise

List five laptop maintenance tasks.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 42: PC Upgrade Budgeting

Lesson Content

A good upgrade balances cost, compatibility, performance improvement, reliability, and future expansion. Upgrading the actual bottleneck usually gives better value.

Exercise

How would you prioritize upgrades for a low-budget PC?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 43: Documentation and Maintenance Records

Lesson Content

Record hardware configuration, firmware versions, installed components, maintenance dates, repairs, tests, and backup status. Documentation supports future troubleshooting.

Exercise

Design a simple PC maintenance record.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 44: Professional Maintenance Workflow

Lesson Content

A professional workflow includes preparation, data protection, diagnosis, authorization where needed, repair or upgrade, testing, documentation, and handover.

Exercise

Describe a professional workflow from complaint to completed repair.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 45: PC Upgrade Case Study

Lesson Content

A school lab has slow startup, limited RAM, old HDDs, overheating cases, and inconsistent software. A phased plan should prioritize backups, storage, RAM where supported, cleaning, cooling, OS standardization, and documentation.

Exercise

Create a phased upgrade plan for the lab.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 46: Final Revision Guide

Lesson Content

Review hardware functions, compatibility, upgrade planning, storage, memory, cooling, firmware, operating systems, preventive maintenance, ESD, troubleshooting, security, backups, and documentation.

Exercise

Explain how you would diagnose a slow PC before buying a component.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Final Assessment: 60 Questions and Answers

Attempt each question before reading the answer.

1. What is PC maintenance?

Answer: Systematic care, inspection, cleaning, updating, troubleshooting, and repair of a computer.

2. What is a PC upgrade?

Answer: Adding or replacing components to improve capability, capacity, compatibility, or performance.

3. Name five major PC components.

Answer: Motherboard, CPU, RAM, storage, and power supply.

4. What does the motherboard do?

Answer: It connects major PC components and provides interfaces for expansion and peripherals.

5. What does the CPU do?

Answer: It executes program instructions and performs calculations.

6. What is RAM?

Answer: Fast temporary memory used by active programs and data.

7. What is an SSD?

Answer: A solid-state storage device using nonvolatile flash memory.

8. What is an HDD?

Answer: A magnetic mechanical storage device using rotating platters.

9. What is NVMe?

Answer: A storage protocol commonly used with PCIe-connected SSDs.

10. Why check compatibility before upgrading?

Answer: To ensure the component fits and works electrically, logically, and thermally with the system.

11. What is a PSU?

Answer: A power supply unit that converts mains electricity into regulated power for components.

12. Why is cooling important?

Answer: It keeps components within suitable temperatures and helps prevent throttling or instability.

13. What is thermal throttling?

Answer: Reducing performance to control excessive temperature.

14. What is thermal interface material?

Answer: Material that improves heat transfer between a processor and cooler.

15. What is BIOS/UEFI?

Answer: Firmware that initializes hardware and provides pre-boot configuration.

16. Why update firmware carefully?

Answer: An incorrect or interrupted update can cause serious system problems.

17. What is preventive maintenance?

Answer: Maintenance intended to reduce future failures.

18. What is corrective maintenance?

Answer: Maintenance performed to fix an existing problem.

19. What is ESD?

Answer: Electrostatic discharge that can damage sensitive electronic components.

20. How can ESD risk be reduced?

Answer: Use suitable anti-static precautions and proper component handling.

21. Why clean a PC?

Answer: To reduce dust and help maintain airflow and cooling.

22. Why is cable management important?

Answer: It improves airflow, organization, and serviceability.

23. What is a driver?

Answer: Software that allows an operating system to communicate with hardware.

24. Why keep software updated?

Answer: Updates can improve security, stability, compatibility, and functionality.

25. What is a backup?

Answer: A separate copy of data used for recovery.

26. Why verify backups?

Answer: A backup that cannot be restored is not dependable.

27. What should be checked before formatting?

Answer: Confirm the correct drive and ensure important data is backed up.

28. What is a file system?

Answer: A method used to organize data on storage.

29. Name three PC file systems.

Answer: NTFS, exFAT, and FAT32.

30. What is troubleshooting?

Answer: A systematic process for identifying, testing, and resolving faults.

31. What should you do before replacing components randomly?

Answer: Measure symptoms and isolate the likely cause.

32. Name three causes of no display.

Answer: Cable/input issues, RAM/GPU problems, and firmware or hardware faults.

33. Name three causes of overheating.

Answer: Dust, failed/inadequate cooling, and poor airflow.

34. Name three causes of a slow PC.

Answer: Insufficient RAM, slow storage, and excessive background activity.

35. What is a UPS?

Answer: An uninterruptible power supply providing temporary backup power and related protection.

36. Why protect against power problems?

Answer: Outages and voltage disturbances can interrupt operation and potentially damage equipment.

37. What should be checked before a CPU upgrade?

Answer: Socket, motherboard firmware, CPU support, power delivery, cooling, and memory compatibility.

38. What should be checked before a GPU upgrade?

Answer: Interface, physical clearance, PSU capacity/connectors, cooling, and software support.

39. What should be checked before a RAM upgrade?

Answer: Memory generation, module type, capacity limits, slots, and supported configurations.

40. Why monitor storage health?

Answer: To identify potential failures early and protect data.

41. Why document maintenance work?

Answer: It provides history and supports future diagnosis.

42. Why use trusted driver sources?

Answer: To reduce incompatible or malicious software risk.

43. What is a clean OS installation?

Answer: Installing an operating system from a clean state.

44. What is disk cloning?

Answer: Creating a copy of a storage drive or its partitions for migration or recovery.

45. What is a bottleneck?

Answer: A subsystem that significantly limits overall performance.

46. Why benchmark before upgrading?

Answer: To establish a baseline and identify actual limitations.

47. What is motherboard form factor?

Answer: A physical standard defining board dimensions and layout.

48. What is a CPU socket?

Answer: The motherboard interface for a compatible processor.

49. Why must a cooler match a CPU socket?

Answer: Its mounting and contact mechanism must fit the processor platform.

50. What is airflow?

Answer: Movement of air through a PC to remove heat.

51. Why should fans not be obstructed?

Answer: Obstruction reduces cooling efficiency.

52. What is corrective repair?

Answer: Work performed to restore a faulty system.

53. Why disconnect power before internal servicing?

Answer: To reduce electrical hazards and accidental damage.

54. Why should components not be forced into connectors?

Answer: Misalignment or force can damage hardware.

55. What is a maintenance schedule?

Answer: A planned timetable for recurring maintenance tasks.

56. What is a service record?

Answer: Documentation of maintenance, repairs, upgrades, tests, and configuration.

57. What is a maintenance checklist?

Answer: A structured list used to ensure important tasks are completed.

58. What is a safe upgrade principle?

Answer: Protect data, verify compatibility, handle parts correctly, install carefully, and test afterward.

Practical Exercises

Exercise 1 — Hardware Identification

Identify the motherboard, CPU cooler, RAM, storage, PSU, expansion slots, fans, and connectors in a desktop PC when it is safe and permitted. Record specifications where available.

Exercise 2 — Upgrade Planning

Choose a PC and identify its CPU, RAM, storage, PSU, motherboard, and cooling. Recommend three useful upgrades and justify them.

Exercise 3 — Maintenance Checklist

Create a monthly checklist covering backups, updates, storage health, cleaning, airflow, cables, security, and performance.

Exercise 4 — Troubleshooting

Create a diagnostic flowchart for a PC that powers on but shows no display. Include monitor, cables, RAM, GPU, firmware, and minimal-hardware tests.

Exercise 5 — Service Report

Create a maintenance report containing complaint, symptoms, diagnosis, work performed, components changed, tests, and final result.

PC Upgrade and Maintenance Glossary

BIOS: Firmware interface used to initialize hardware before OS startup.

UEFI: Modern firmware environment for hardware initialization and boot configuration.

CPU: Central Processing Unit.

GPU: Graphics Processing Unit.

RAM: Random Access Memory.

SSD: Solid-state storage device.

HDD: Magnetic mechanical storage device.

NVMe: Protocol commonly used for PCIe solid-state storage.

PSU: Power Supply Unit.

Motherboard: Main circuit board connecting PC components.

Firmware: Low-level software stored on hardware.

Driver: Software enabling OS-to-hardware communication.

Thermal throttling: Reducing performance to control excessive temperature.

ESD: Electrostatic discharge.

Backup: Separate copy of data for recovery.

UPS: Uninterruptible Power Supply.

Bottleneck: Subsystem limiting overall performance.

Form factor: Physical standard defining size and layout.

File system: Method for organizing data on storage.

Preventive maintenance: Maintenance intended to reduce future failures.

Corrective maintenance: Maintenance performed to fix an existing fault.

End of PC Upgrade and Maintenance

Safe and effective PC servicing depends on compatibility verification, data protection, controlled troubleshooting, correct handling, proper cooling, reliable power, and thorough documentation.